

Activated-Range Represented Grams at a Fixed Shift: Repair Invariance, Multiblock Kernel Fusion, and the Failure of Row Capacity Transfer

Liang Wang

School of Artificial Intelligence and Automation,
Huazhong University of Science and Technology, Wuhan 430074, P.R. China
wangliang.f@gmail.com

July 2026

Abstract

The terminal-activation calculus for a fixed-shift multiscale incidence leaves one precise question: how much of the activated energy survives the represented output map? We answer the finite-dimensional part on the *actual range* generated by the physical row coefficients. The optimal represented-to-terminal constant is the smallest eigenvalue of an activated row Gram; in the non-descending case it is instead the shorted generalized Gram already required by the null directions. We give an explicit residual-fiber formula for every matrix entry and prove that terminal-faithful pair repair leaves the activated Gram exactly unchanged whenever the downstream map factors through the literal terminal carrier. Orthogonally labelled blocks satisfy a kernel-fusion law: each block may be singular, while their sum is positive precisely when their kernels have trivial intersection. A weighted support-projection bound turns this into a sharp spectral-capacity certificate. The main obstruction is also exact: rowwise activation capacities do not transfer through a coherent represented map unless a reducing or basis-transport theorem is supplied. Thus parent inflation is charged once at activation, whereas coherent row collision is a separate loss. All results are exact finite-scale statements. They identify the next arithmetic task—control of the actual residual-fiber column geometry—but prove neither a polynomial physical lower bound nor a prime-pair conclusion.

Keywords: fixed shift; terminal activation; represented Gram; shorted operator; spectral capacity; coherent collision.

Convention. Every arithmetic specialization uses one fixed nonzero shift h_0 , literal physical coefficients, actual active support, and raw counting-measure atomic norms. The shift is not specialized to 2. No finite-dimensional Gram identity is promoted to a parity estimate or a prime-pair lower bound.

Inner-product convention. Inner products are conjugate linear in the first variable and linear in the second.

MSC 2020: Primary 15A18, 47A64; Secondary 11N35, 11P32.

1 Introduction

The preceding provenance and activation papers separated three objects that are easily conflated: a declared parent incidence, the literal terminal atoms, and the energy that is actually generated by a fixed row coefficient [4, 5]. That separation resolves which denominator is being charged. It does not show that a coherent residual or native representation preserves the activated energy. The latter is a matrix question on the range populated by the physical coefficients.

This paper closes that interface at the exact finite-dimensional level. Its first point is a restriction principle: the free ambient terminal atom space is not the represented reference. The correct carrier is the actual analysis range, or equivalently the original row space equipped with the semidefinite terminal form. On this carrier the represented map has a concrete column Gram. Large kernels of an ambient grouping map can be irrelevant when they miss the populated range; conversely, every individual row can be nonzero while two active row columns cancel coherently.

The general shorted-operator formula is not new here. It was proved and audited in the exact closure calculus of Wang [3], building on the classical shorted-operator viewpoint [1]. We use that formula only to distinguish two cases. If the represented map descends through terminalization, its terminal-null directions vanish and the short reduces to an ordinary Gram on the actual range. If it does not descend, deleting those directions would be an invalid compression; the full short must be retained.

The new content is fourfold.

- (i) We derive the literal residual-fiber row Gram and an exact criterion for represented positivity on the activated range.
- (ii) We prove invariance of this Gram under every terminal-faithful pair repair. Repair-only parent atoms affect activation relative to the parent but cannot be charged again after terminalization.
- (iii) We establish a multiblock kernel-fusion bound. Singular dyadic block Grams can combine to a positive global Gram when their blind directions are complementary.
- (iv) We prove a sharp spectral-capacity law and an equally sharp non-transfer theorem: the row-capacity certificate from the activation layer survives only after a reducing/basis-transport statement connects the row basis to the good spectral subspace.

These conclusions advance the interface but not the arithmetic exponent. In particular, injectivity at each finite X need not give a polynomial lower bound, and a labelled isometry need not survive physical label erasure. The next step must therefore study the literal collision geometry of the residual-fiber columns, rather than infer it from activation counts.

Claim level and quantifiers

The linear-algebraic results are unconditional L0 statements for the declared finite spaces and maps. Their fixed- h_0 dictionary is L1 only when the same literal coefficients, actual support, block labels, and raw atomic norm from the activation paper are retained. No result below proves a favorable activation census, a polynomial lower singular value for the physical unlabelled map, a parity-breaking estimate, or a prime-pair lower bound.

2 The literal activated range

Fix a scale X , a nonzero shift h_0 , and every external choice before the coefficient vector. Let

$$Z_X = \mathbb{C}^{\mathcal{M}_X}, \quad z = (z_m)_{m \in \mathcal{M}_X},$$

be the physical row space. The terminal construction of Wang [5] gives a finite Hilbert space

$$\mathcal{H}_X^{\text{term}} = \bigoplus_{k \in K_X}^{\perp} \ell^2(\mathcal{U}_{X,h_0,k}^+)$$

and one literal analysis map

$$L_X : Z_X \longrightarrow \mathcal{H}_X^{\text{term}}.$$

Its columns are

$$f_m = L_X e_m, \quad T_m = \|f_m\|^2.$$

Different physical rows populate disjoint labelled terminal coordinates at this stage, so

$$T_X := L_X^* L_X = \text{diag}(T_m), \quad \mathcal{T}_X(z) = \|L_X z\|^2 = \sum_m T_m |z_m|^2. \quad (1)$$

The *activated range* is

$$Y_X := \text{Ran } L_X \subseteq \mathcal{H}_X^{\text{term}}.$$

It is this range, not the free ambient terminal space, that enters every represented-reference assertion below.

TPC-64 supplies two compatible parent forms. Write

$$R_{X,\text{hyp}} = \text{diag}(P_m), \quad R_{X,\text{rep}} = \text{diag}(\tilde{P}_m),$$

for the literal-hypergraph and canonical-repair forms. Thus

$$\mathcal{P}_{X,\nu}(z) = \langle R_{X,\nu} z, z \rangle, \quad \nu \in \{\text{hyp}, \text{rep}\}.$$

When the finite inflation constant $\mathfrak{I}_X$ is available,

$$R_{X,\text{hyp}} \preceq R_{X,\text{rep}} \preceq \mathfrak{I}_X R_{X,\text{hyp}}. \quad (2)$$

Definition 2.1 (Descending map). A row-space map $F_X : Z_X \rightarrow H_X$ *descends through terminalization* if there is a linear map $A_X : Y_X \rightarrow H_X$ such that

$$F_X = A_X L_X.$$

Equivalently, $\text{Ker } L_X \subseteq \text{Ker } F_X$.

Lemma 2.2 (Exact quotient/range criterion). *A linear map $F_X : Z_X \rightarrow H_X$ descends through L_X if and only if $\text{Ker } L_X \subseteq \text{Ker } F_X$. When it descends, the induced map on Y_X is unique.*

Proof. Necessity is immediate. Conversely, define $A_X(L_X z) = F_X z$. If $L_X z = L_X z'$, then $z - z' \in \text{Ker } L_X$, so $F_X z = F_X z'$. The definition is well posed, linear, and unique because $L_X Z_X = Y_X$. $\square$

The distinction is essential. A map acting before terminalization may read repair-only parent coordinates which terminalize to zero. Such a map need not descend, and neither the activated-range Gram nor the repair invariance proved later applies to it.

2.1 Activation kernels and transported parent metrics

Let

$$S_T = (\text{Ker } T_X)^\perp = \text{span}\{e_m : T_m > 0\}.$$

The restriction $L_X|_{S_T} : S_T \rightarrow Y_X$ is an isomorphism. If, for a parent form $R_{X,\nu}$,

$$\text{Ker } R_{X,\nu} = \text{Ker } T_X, \quad (3)$$

then the parent metric transports to Y_X by

$$\hat{R}_{X,\nu} = (L_X|_{S_T}^{-1})^* R_{X,\nu}|_{S_T} (L_X|_{S_T}^{-1}).$$

In the normalized row basis $u_m = f_m/\sqrt{T_m}$, this metric is diagonal:

$$[\hat{R}_{X,\nu}]_{u_m} = \text{diag} \left(\frac{R_{X,\nu}(m, m)}{T_m} \right)_{T_m > 0}. \quad (4)$$

This is a transported *parent* metric, not the represented Gram.

Proposition 2.3 (Mandatory activation kill). *If there is $z \in \text{Ker } L_X$ with $\langle R_{X,\nu} z, z \rangle > 0$, then for every downstream map A_X defined on Y_X ,*

$$\inf_{\mathcal{P}_{X,\nu}(w) > 0} \frac{\|A_X L_X w\|^2}{\mathcal{P}_{X,\nu}(w)} = 0.$$

For the diagonal forms this occurs exactly when some row has $R_{X,\nu}(m, m) > 0 = T_m$.

Proof. Use $w = z$. The numerator vanishes and the denominator is positive. The diagonal characterization follows from (1). $\square$

Thus the kernel equality (3) is not cosmetic. Failure of the reverse inclusion is an activation-layer obstruction that no later represented operator can repair.

3 The activated row Gram

Assume $Y_X \neq \{0\}$. Let $A_X : Y_X \rightarrow H_X^{\text{rep}}$ be a declared represented map and set

$$F_X = A_X L_X : Z_X \rightarrow H_X^{\text{rep}}, \quad G_X = F_X^* F_X.$$

The exact represented-to-terminal constant is

$$\beta_X := \inf_{\mathcal{T}_X(z) > 0} \frac{\|F_X z\|^2}{\mathcal{T}_X(z)}. \quad (5)$$

Theorem 3.1 (Activated-range represented Gram). *On $S_T = (\text{Ker } T_X)^\perp$, define*

$$K_X = T_{X,S}^{-1/2} G_{X,S} T_{X,S}^{-1/2}. \quad (6)$$

Then

$$\boxed{\beta_X = \lambda_{\min}(K_X) = \sigma_{\min}(A_X|_{Y_X})^2.} \quad (7)$$

Consequently the following are equivalent:

- (i) $\beta_X > 0$;
- (ii) $\text{Ker } A_X \cap Y_X = \{0\}$;
- (iii) $\text{Ker } F_X = \text{Ker } L_X$;
- (iv) *the represented columns $A_X f_m$, $T_m > 0$, are linearly independent.*

In particular, $\dim Y_X > \dim H_X^{\text{rep}}$ forces $\beta_X = 0$.

Proof. Every $z = z_S + z_0$ with $z_0 \in \text{Ker } T_X = \text{Ker } L_X$ has the same numerator and denominator as z_S . With $x = T_{X,S}^{1/2} z_S$, the quotient in (5) becomes $\langle K_X x, x \rangle / \|x\|^2$. This proves the first equality. The map $U = L_X T_{X,S}^{-1/2} : S_T \rightarrow Y_X$ is unitary because $U^* U = I$, and $K_X = U^* A_X^* A_X U$, proving the singular-value identity. The kernel equivalences and dimension statement follow. $\square$

3.1 Literal residual-fiber formula

Suppose the represented output has an orthonormal coordinate set $\mathcal{Y}_X$, and terminal atoms u are sent to output fibers $q(u) \in \mathcal{Y}_X$. Write the terminal coefficient contributed by row m at atom u as a_u . Thus, for a terminal atom generated by row $m(u)$, its coefficient in $L_X z$ is $a_u z_{m(u)}$. Define the coherent row-fiber column

$$b_{y,m} := \sum_{\substack{u: q(u)=y \\ m(u)=m}} a_u. \quad (8)$$

Then

$$(F_X z)_y = \sum_m b_{y,m} z_m, \quad (9)$$

and hence

$$(G_X)_{m,n} = \sum_{y \in \mathcal{Y}_X} \overline{b_{y,m}} b_{y,n}. \quad (10)$$

Formula (10) is the concrete arithmetic target. Diagonal entries measure represented row energy, while off-diagonal entries record actual coherent collisions. An ambient atom-grouping operator can have a large kernel without harming K_X if that kernel misses Y_X . Conversely, nonzero diagonal entries do not prevent a kernel among the row columns.

Corollary 3.2 (Common output gauge cancels). *If $b_{y,m} = \Omega_y \beta_{y,m}$, where Ω_y is independent of m , then*

$$(G_X)_{m,n} = \sum_y |\Omega_y|^2 \overline{\beta_{y,m}} \beta_{y,n}.$$

In particular, a common unit-modulus Möbius or phase gauge on an output fiber cannot by itself improve or degrade the represented Gram.

3.2 When the map does not descend

For completeness, let $\tilde{F}_X : Z_X \rightarrow H_X$ be a row-space map not known to kill $\text{Ker } T_X$. Ordinary compression to S_T is then invalid. The exact constant is the TPC-62 shorted Gram

$$\alpha_{T_X}(\tilde{F}_X) = \lambda_{\min} \left(T_{X,S}^{-1/2} \tilde{F}_{X,S}^* P_{\tilde{F}_X(\text{Ker } T_X)^\perp} \tilde{F}_{X,S} T_{X,S}^{-1/2} \right), \quad (11)$$

not the unprojected compression [3]. Equation (6) is valid here precisely because the declared map is $A_X L_X$ and therefore annihilates $\text{Ker } T_X$.

4 Labelled kernel fusion and coherent label erasure

The external multiscale label k is retained orthogonally in the literal terminal carrier. Assume $Y_X \neq \{0\}$. This creates two distinct represented maps which must not be conflated.

Let $F_{X,k} : Z_X \rightarrow H_{X,k}$ be the represented map from block k . Assume throughout this section that

$$\text{Ker } T_X = \text{Ker } L_X \subseteq \text{Ker } F_{X,k} \quad (k \in K_X), \quad (12)$$

so every block descends through the activated quotient. Define the labelled direct sum

$$F_X^{\text{lab}} z = \bigoplus_{k \in K_X} F_{X,k} z.$$

Put

$$G_{X,k} = F_{X,k}^* F_{X,k}, \quad K_{X,k} = T_{X,S}^{-1/2} G_{X,k} T_{X,S}^{-1/2}.$$

Its exact labelled lower constant is

$$\beta_X^{\text{lab}} := \inf_{\tau_X(z) > 0} \frac{\sum_k \|F_{X,k}z\|^2}{\tau_X(z)}. \quad (13)$$

Theorem 4.1 (Multiblock kernel fusion). *The labelled normalized Gram and its lower constant are*

$$K_X^{\text{lab}} = \sum_k K_{X,k}, \quad \beta_X^{\text{lab}} = \lambda_{\min} \left(\sum_k K_{X,k} \right). \quad (14)$$

In particular,

$$\beta_X^{\text{lab}} > 0 \iff \bigcap_k \text{Ker } K_{X,k} = \{0\} \quad \text{on } S_T. \quad (15)$$

For each nonzero $K_{X,k}$, let P_k be its support projection and $\eta_k = \lambda_{\min}^+(K_{X,k})$. Then

$$\boxed{\beta_X^{\text{lab}} \geq \lambda_{\min} \left(\sum_k \eta_k P_k \right)}. \quad (16)$$

Thus individually singular blocks can produce a positive labelled Gram when their blind subspaces are complementary.

Proof. Orthogonality of the output labels gives $(F_X^{\text{lab}})^* F_X^{\text{lab}} = \sum_k G_{X,k}$, proving (14). For positive semidefinite matrices, $\text{Ker}(\sum_k K_{X,k}) = \bigcap_k \text{Ker } K_{X,k}$, which proves (15). The spectral theorem gives $K_{X,k} \succeq \eta_k P_k$; summing and taking the smallest eigenvalue gives (16). $\square$

4.1 The physical coherent map

Let $U_k : H_{X,k} \rightarrow H_X^{\text{phys}}$ be the maps that erase labels or place block outputs in one physical codomain. The coherent represented map is

$$F_X^{\text{coh}} z = \sum_k U_k F_{X,k} z.$$

Its full Gram is

$$G_X^{\text{coh}} = \sum_{k,\ell} F_{X,k}^* U_k^* U_\ell F_{X,\ell}. \quad (17)$$

The cross terms in (17) are physical. They cannot be discarded merely because the input construction once carried orthogonal labels. Define the coherent lower constant by

$$\beta_X^{\text{coh}} := \inf_{\tau_X(z) > 0} \frac{\|F_X^{\text{coh}} z\|^2}{\tau_X(z)}. \quad (18)$$

Proposition 4.2 (Fusion-angle transfer). *Assume that for every family $v_k \in F_{X,k}(Z_X)$,*

$$\left\| \sum_k U_k v_k \right\|^2 \geq \theta_X \sum_k \|v_k\|^2 \quad (19)$$

with $\theta_X \geq 0$. Then

$$\beta_X^{\text{coh}} \geq \theta_X \beta_X^{\text{lab}}.$$

This implication is sharp already in one dimension.

Proof. Apply (19) to $v_k = F_{X,k}z$, then use the definition of β_X^{lab} . Equality is attained when both inequalities are equalities on the same vector. For an explicit equality model take one block with $T = I$, $F = I$, and $U = \sqrt{\theta_X}I$. $\square$

No positive θ_X follows from the individual block lower bounds. For $F_1 = F_2 = I$, $U_1 = I$, and $U_2 = -I$, each block is an isometry while $F^{\text{coh}} = 0$. The labelled theorem is therefore a genuine positive mechanism, but the physical unlabelled route additionally requires a fusion-angle or cross-Gram estimate. Identifying the literal fixed-shift map and its active support is L1; proving an endpoint-quality physical estimate is an L2 arithmetic task. Neither follows from multiscale bookkeeping.

5 Spectral capacity after exact nuisance elimination

The general shorted-Gram formula and its kernel criterion belong to the closure calculus developed in TPC-62; we do not reprove them here. We begin after exact nuisance elimination has produced a positive semidefinite effective Gram Q on the reference-visible space S , together with a positive definite reference Gram $R_{\text{ref},S}$. In normalized coordinates,

$$K_{\text{eff}} := R_{\text{ref},S}^{-1/2} Q R_{\text{ref},S}^{-1/2} \geq 0, \quad x := R_{\text{ref},S}^{1/2} s, \quad (20)$$

the exact short implies that every represented/reference quotient with visible coordinate s is bounded below by

$$\frac{\langle K_{\text{eff}} x, x \rangle}{\|x\|^2}. \quad (21)$$

Thus the remaining question is finite-dimensional: when do structural capacity constraints force a vector to place a definite fraction of its energy in a spectrally good part of a fixed positive semidefinite Gram?

5.1 Reducing blocks and structural spectral capacity

Definition 5.1 (Spectral-capacity data). Let $\mathcal{S}$ be a finite-dimensional Hilbert space, let $K: \mathcal{S} \rightarrow \mathcal{S}$ be positive semidefinite, and fix an orthogonal K -reducing decomposition over a finite index set $\mathcal{J}$, with every $\mathcal{S}_j \neq \{0\}$:

$$\mathcal{S} = \bigoplus_{j \in \mathcal{J}} \mathcal{S}_j, \quad K P_j = P_j K, \quad (22)$$

where P_j is the orthogonal projection onto $\mathcal{S}_j$. Assign a structural capacity weight $w_j > 0$ to each block and put

$$0 < W := \sum_{j \in \mathcal{J}} w_j < \infty.$$

These weights are declared energy budgets; they need not equal block dimensions, ranks, traces, or spectral multiplicities.

For a threshold $\eta > 0$, define

$$\mathcal{G}_\eta := \left\{ j \in \mathcal{J} : K|_{\mathcal{S}_j} \geq \eta I_{\mathcal{S}_j} \right\}, \quad \delta_\eta := \frac{\sum_{j \in \mathcal{G}_\eta} w_j}{W}. \quad (23)$$

Given $C > 0$ and $0 < \kappa \leq 1$, a nonzero vector $x \in \mathcal{S}$ belongs to the corresponding spectral-capacity class if

$$\|P_j x\|^2 \leq C^2 w_j \quad (j \in \mathcal{J}), \quad \|x\|^2 \geq \kappa C^2 W. \quad (24)$$

Theorem 5.2 (Sharp spectral-capacity lower bound). *For the data of [definition 5.1](#), every vector satisfying (24) obeys*

$$\boxed{\frac{\langle Kx, x \rangle}{\|x\|^2} \geq \eta \left(1 - \frac{1 - \delta_\eta}{\kappa} \right)_+}, \quad t_+ := \max\{t, 0\}. \quad (25)$$

In particular, the right-hand side is positive precisely when

$$\kappa > 1 - \delta_\eta. \quad (26)$$

The coefficient in (25) is optimal if one knows only η , δ_η , κ , the block caps, and the total occupancy condition.

Proof. Orthogonality and the block caps give

$$\sum_{j \notin \mathcal{G}_\eta} \|P_j x\|^2 \leq C^2 \sum_{j \notin \mathcal{G}_\eta} w_j = C^2(1 - \delta_\eta)W. \quad (27)$$

Dividing this estimate by $\|x\|^2 \geq \kappa C^2 W$ shows that the fraction of the energy in the good blocks is at least

$$\frac{\sum_{j \in \mathcal{G}_\eta} \|P_j x\|^2}{\|x\|^2} \geq \left(1 - \frac{1 - \delta_\eta}{\kappa}\right)_+.$$

Because the decomposition is K -reducing, there are no cross terms between distinct blocks. Positivity of K on the remaining blocks and the definition of $\mathcal{G}_\eta$ therefore yield

$$\langle Kx, x \rangle = \sum_{j \in \mathcal{J}} \langle KP_j x, P_j x \rangle \geq \eta \sum_{j \in \mathcal{G}_\eta} \|P_j x\|^2.$$

This proves (25), and (26) follows immediately.

It remains to prove optimality. Normalize $C = W = 1$. First suppose $0 < \delta_\eta < 1$, and take two one-dimensional reducing blocks with weights

$$w_g = \delta_\eta, \quad w_b = 1 - \delta_\eta,$$

on which K acts by η and 0, respectively. If $\kappa \leq 1 - \delta_\eta$, choose

$$\|x_g\|^2 = 0, \quad \|x_b\|^2 = \kappa.$$

All caps hold, the total energy is κ , and the Rayleigh quotient is zero. If $\kappa > 1 - \delta_\eta$, choose instead

$$\|x_b\|^2 = 1 - \delta_\eta, \quad \|x_g\|^2 = \kappa - 1 + \delta_\eta.$$

The good-block cap holds because $\kappa \leq 1$, and the resulting quotient is

$$\eta \frac{\kappa - 1 + \delta_\eta}{\kappa} = \eta \left(1 - \frac{1 - \delta_\eta}{\kappa}\right).$$

For the endpoint $\delta_\eta = 0$, a single bad block with energy κ gives quotient zero. For $\delta_\eta = 1$, a single good block on which $K = \eta I$, again with energy κ , gives quotient η . Thus equality is attained in every parameter regime. $\square$

Remark 5.3 (Actual occupancy). For a fixed capped vector $x \neq 0$ and a fixed cap scale C , the actual occupancy is

$$\kappa(x; C) := \frac{\|x\|^2}{C^2 W} \in (0, 1]. \quad (28)$$

Applying [theorem 5.2](#) with this value, rather than with a weaker a priori lower bound, gives

$$\frac{\langle Kx, x \rangle}{\|x\|^2} \geq \eta \left(1 - \frac{1 - \delta_\eta}{\kappa(x; C)}\right)_+.$$

Combining this estimate with (21) gives the exact capacity consequence for the original quotient. It applies once the normalized effective Gram and a valid reducing capacity ledger have both been identified.

5.2 Why physical row capacity does not transfer

The reducing qualification in [definition 5.1](#) is indispensable. Diagonal information in a physical row basis does not control coherent off-diagonal cancellation.

Proposition 5.4 (Sharp row-capacity non-transfer). *There is a positive semidefinite Gram for which every physical coordinate row has unit diagonal gain and a vector has maximal row-capacity occupancy, while the represented energy of that vector is zero. Consequently, physical row-capacity data alone imply no positive spectral-capacity lower bound.*

Proof. Let

$$\mathcal{S} = \mathbb{C}^2, \quad A = \begin{pmatrix} 1 & -1 \end{pmatrix}, \quad K = A^* A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}. \quad (29)$$

For the physical coordinate vectors e_1, e_2 ,

$$\langle K e_1, e_1 \rangle = \langle K e_2, e_2 \rangle = 1.$$

Hence a ledger that treats the two physical coordinate rows as independent good blocks at threshold $\eta = 1$ would record $\delta_{\text{row}} = 1$. Give the rows weights $w_1 = w_2 = 1$, take $C = 1$, and set

$$x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Both row caps are saturated, and the row occupancy is maximal:

$$\kappa_{\text{row}} = \frac{\|x\|^2}{C^2(w_1 + w_2)} = 1.$$

Nevertheless,

$$Ax = 0, \quad \langle Kx, x \rangle = 0.$$

The invalid scalar substitution $(\eta, \delta, \kappa) = (1, 1, 1)$ into [\(25\)](#) would predict the lower bound 1, whereas the true Rayleigh quotient is 0.

The spectral resolution explains the failure. The eigenvalues of K are 0 and 2, and its kernel is the coherent direction $\text{span}\{(1, 1)\}$. The physical coordinate decomposition $\mathbb{C}e_1 \oplus \mathbb{C}e_2$ does not reduce K , so it is not an admissible decomposition in [definition 5.1](#). The off-diagonal terms cancel two individually favorable diagonal gains. Both diagonal gains and the occupancy saturate the chosen unit normalization, so no strengthening based only on those scalar row data can repair the conclusion. $\square$

Remark 5.5 (Valid recovery routes). A physical row-capacity certificate can be transferred to [theorem 5.2](#) only after an additional structural statement is proved. Sufficient routes include:

1. the physical one-row decomposition itself reduces K (equivalently, the relevant Gram is diagonal), or an explicitly declared row grouping gives a reducing block-diagonal decomposition;
2. a quantitative basis-transport or coherence estimate converts the physical row caps into caps for a declared K -reducing decomposition; or
3. the argument is formulated directly with the spectral projectors of the exact effective Gram.

Without one of these bridges, no favorable capacity conclusion is asserted.

5.3 Claim boundary

[Theorem 5.2](#) and [proposition 5.4](#) are finite-dimensional L0 statements. The operator K , its reducing blocks, the weights, and the capacity caps are declared data. This section does not prove that a physical TPC represented Gram has a favorable value of δ_η , a favorable occupancy κ , or a positive smallest eigenvalue. It makes no L1 arithmetic identification, does not specialize to the fixed shift $h_0 = 2$, and yields no parity-breaking, prime-pair, or twin-prime conclusion.

6 Terminal-faithful repair invariance and parent covariance

This section compares the literal-active-hypergraph carrier with the canonical pair-repaired carrier. The comparison is made after the typed terminal operator, where both constructions take values in the same literal terminal Hilbert space. This placement is essential: a map acting on a pre-terminal parent space may see repair-only coordinates which the terminal operator annihilates.

6.1 One terminal map from two parent carriers

Retain the finite scale X , the fixed nonzero shift h_0 , and the coefficient space

$$Z_X = \mathbb{C}^{\mathcal{M}_X}.$$

In the notation of Wang [\[5\]](#), define

$$L_X^{\text{hyp}} := D_{X,h_0}^{\mathcal{U}^{\text{pre},+}} C_X^{\text{hyp}}, \quad (30)$$

$$L_X^{\text{rep}} := D_{X,h_0}^{\mathcal{V}^{\text{hyb}}} C_X^{\text{hyb}}. \quad (31)$$

Both maps have codomain

$$\mathcal{H}_X^{\text{term}} = \bigoplus_{k \in K_X}^{\perp} \ell^2(\mathcal{U}_{X,h_0,k}^+),$$

the literal terminal atom space. Their parent spaces are different, but their displayed codomain is not.

Theorem 6.1 (Common terminal map). *The hypergraph and canonical pair-repaired constructions induce the same terminal map:*

$$\boxed{L_X^{\text{hyp}} = L_X^{\text{rep}} =: L_X.} \quad (32)$$

Moreover,

$$\boxed{L_X^* L_X = T_X = \text{diag}(T_m).} \quad (33)$$

Consequently their activated ranges agree exactly:

$$Y_X := \text{Ran } L_X^{\text{hyp}} = \text{Ran } L_X^{\text{rep}} = \text{Ran } L_X. \quad (34)$$

Proof. Fix a literal terminal coordinate (k, u) , with $u \in \mathcal{U}_{X,h_0,k}^+$. Both populated parent analyses assign the value $c_u z_{m(u)}$ to this coordinate, and both typed terminal operators multiply it by the same literal multiplier $t_{h_0}(u)$. Thus

$$(L_X^{\text{hyp}} z)(k, u) = t_{h_0}(u) c_u z_{m(u)} = (L_X^{\text{rep}} z)(k, u)$$

for every $z \in Z_X$. The typed repaired terminal operator discards every repair-only coordinate before entering the displayed literal terminal codomain, so those coordinates contribute nothing. This proves [\(32\)](#).

Distinct row indices populate disjoint labelled terminal coordinates. Hence the columns $L_X e_m$ are mutually orthogonal, and their squared norms are precisely

$$\|L_X e_m\|^2 = \sum_{k \in K_X} \sum_{\substack{u \in \mathcal{U}_{X, h_0, k}^+ \\ m(u)=m}} |t_{h_0}(u) c_u|^2 = T_m.$$

This proves (33), and (34) follows immediately. $\square$

6.2 The exact represented Gram on the common range

Let

$$A_X : Y_X \longrightarrow \mathcal{H}_X^{\text{rep}}$$

be one declared represented map on the common activated range. This means that the hypergraph and repaired descriptions use the same downstream map after terminalization. Assume $Y_X \neq \{0\}$; when $Y_X = \{0\}$, there is no nonzero represented reference and no lower constant is assigned. Define

$$\beta_X := \inf_{0 \neq y \in Y_X} \frac{\|A_X y\|^2}{\|y\|^2}. \quad (35)$$

Put

$$\mathcal{M}_X^T := \{m \in \mathcal{M}_X : T_m > 0\}, \quad u_m := \frac{L_X e_m}{\sqrt{T_m}} \quad (m \in \mathcal{M}_X^T).$$

By (33), $(u_m)_{m \in \mathcal{M}_X^T}$ is an orthonormal basis of Y_X . Define its represented row Gram by

$$\mathbf{G}_X^{\text{act}}(m, n) := \langle A_X u_m, A_X u_n \rangle, \quad m, n \in \mathcal{M}_X^T. \quad (36)$$

Proposition 6.2 (Exact and repair-invariant represented constant). *One has*

$$\boxed{\beta_X = \lambda_{\min}(\mathbf{G}_X^{\text{act}}) = \sigma_{\min}(A_X|_{Y_X})^2.} \quad (37)$$

In particular,

$$\boxed{\beta_X > 0 \iff \text{Ker } A_X \cap Y_X = \{0\}.} \quad (38)$$

The matrix $\mathbf{G}_X^{\text{act}}$, the number β_X , and the kernel test are identical for the hypergraph and canonical repaired descriptions.

Proof. The matrix in (36) is the matrix of $A_X^* A_X|_{Y_X}$ in the orthonormal basis (u_m) . The Rayleigh principle therefore gives

$$\inf_{0 \neq y \in Y_X} \frac{\langle A_X^* A_X y, y \rangle}{\|y\|^2} = \lambda_{\min}(\mathbf{G}_X^{\text{act}}),$$

which is (37). In finite dimensions the smallest eigenvalue is positive exactly when the restricted map is injective, proving (38). Repair invariance then follows from [theorem 6.1](#): both constructions have the same L_X , the same range Y_X , the same normalized row basis, and the same declared map A_X . $\square$

Remark 6.3 (What is and is not invariant). The repair-invariant object is the represented Gram on Y_X . The diagonal matrix with entries P_m/T_m , when it is defined, is instead the literal parent metric transported to Y_X ; replacing P_m by $\tilde{P}_m$ changes that metric. It must not be called the represented Gram.

6.3 Direct closure relative to the two parents

Let

$$R_{X,\text{hyp}} := \text{diag}(P_m), \quad R_{X,\text{rep}} := \text{diag}(\tilde{P}_m). \quad (39)$$

For the canonical hybrid repair there are no new-only rows, so, provided the literal parent is nonzero,

$$R_{X,\text{hyp}} \preceq R_{X,\text{rep}} \preceq \mathfrak{I}_X R_{X,\text{hyp}}, \quad 1 \leq \mathfrak{I}_X < \infty, \quad (40)$$

and the two parent operators have the same kernel. Let

$$B_X : Y_X \longrightarrow \mathcal{H}_X^{\text{out}}$$

be any fixed downstream map; it may be the represented map A_X or a complete represented-plus-missing output. For $\nu \in \{\text{hyp}, \text{rep}\}$, define the direct parent-to-output constant

$$\Gamma_{X,\nu}(B_X) := \inf_{\langle R_{X,\nu} z, z \rangle > 0} \frac{\|B_X L_X z\|^2}{\langle R_{X,\nu} z, z \rangle}. \quad (41)$$

Theorem 6.4 (Exact direct Gram and repair covariance). *Let*

$$N_X := \text{Ker } R_{X,\text{hyp}} = \text{Ker } R_{X,\text{rep}}, \quad S_X := N_X^\perp.$$

Then $N_X \subseteq \text{Ker } L_X$, and for each $\nu \in \{\text{hyp}, \text{rep}\}$,

$$\boxed{\Gamma_{X,\nu}(B_X) = \lambda_{\min} \left(R_{X,\nu,S}^{-1/2} L_{X,S}^* B_X^* B_X L_{X,S} R_{X,\nu,S}^{-1/2} \right)}. \quad (42)$$

Its exact positivity test is

$$\boxed{\Gamma_{X,\nu}(B_X) > 0 \iff \text{Ker}(B_X L_X) \subseteq \text{Ker } R_{X,\nu}}. \quad (43)$$

The two parent normalizations obey the sharp sandwich

$$\boxed{\frac{1}{\mathfrak{I}_X} \Gamma_{X,\text{hyp}}(B_X) \leq \Gamma_{X,\text{rep}}(B_X) \leq \Gamma_{X,\text{hyp}}(B_X)}. \quad (44)$$

Proof. The literal implication $P_m = 0 \Rightarrow T_m = 0$, together with (33), gives $N_X \subseteq \text{Ker } L_X$. Consequently $B_X L_X$ also annihilates N_X . Every numerator and denominator in (41) therefore depends only on the S_X -component of z . On S_X , the parent operator is positive definite. The substitution

$$x = R_{X,\nu,S}^{1/2} z$$

turns the quotient into the Rayleigh quotient of the matrix in (42). This proves the exact formula. Its smallest eigenvalue is positive precisely when $B_X L_X$ has no kernel vector outside N_X , giving (43). This is the compatible-kernel case of the general shorted formula of Wang [3]; no nuisance-output projection remains.

The common kernel in (40) makes the admissible set of nonzero denominators the same for both parents. For every vector in this set,

$$\frac{1}{\mathfrak{I}_X} \frac{\|B_X L_X z\|^2}{\langle R_{X,\text{hyp}} z, z \rangle} \leq \frac{\|B_X L_X z\|^2}{\langle R_{X,\text{rep}} z, z \rangle} \leq \frac{\|B_X L_X z\|^2}{\langle R_{X,\text{hyp}} z, z \rangle}.$$

Taking infima proves (44). □

Example 6.5 (Sharpness of the repair charge). Take $Z_X = Y_X = \mathcal{H}_X^{\text{out}} = \mathbb{C}$, $L_X z = \sqrt{a} z$, $B_X = I$, $R_{X,\text{hyp}} = 1$, and $R_{X,\text{rep}} = \mathfrak{I}_X$, where $a > 0$ and $\mathfrak{I}_X \geq 1$. Then

$$\Gamma_{X,\text{hyp}} = a, \quad \Gamma_{X,\text{rep}} = \frac{a}{\mathfrak{I}_X}.$$

Thus the factor $1/\mathfrak{I}_X$ cannot be improved. The coefficient one in the upper inequality is attained when the two parent forms agree globally, in particular when $\mathfrak{I}_X = 1$.

Define the activation constants

$$\tau_{X,\nu} := \inf_{\langle R_{X,\nu} z, z \rangle > 0} \frac{\|L_X z\|^2}{\langle R_{X,\nu} z, z \rangle}. \quad (45)$$

Applying (35) to $y = L_X z$ gives the useful but generally non-exact staged certificate

$$\Gamma_{X,\nu}(A_X) \geq \beta_X \tau_{X,\nu}. \quad (46)$$

Here β_X is repair invariant; the parent repair is charged through $\tau_{X,\nu}$, or equivalently through the direct sandwich (44). Charging $\mathfrak{I}_X$ once in activation and again in the represented Gram would double-count the same parent enlargement.

Remark 6.6 (Generic repairs with new-only rows). The finite lower comparison in (44) relies on the absence of a row with $P_m = 0 < \tilde{P}_m$. For a generic larger repair such a row has $T_m = 0$, hence $L_X e_m = 0$, while its repaired-parent energy is positive. It follows directly from (41) that $\Gamma_{X,\text{rep}}(B_X) = 0$ for every downstream B_X . Thus terminal faithfulness alone does not provide a finite repair charge; the no-new-only property of the canonical hybrid carrier is substantive.

6.4 Why pre-terminal maps are not repair invariant

Example 6.7 (A terminal-null coordinate read before terminalization). Let $Z = \mathbb{C}$ and embed both parent descriptions in $\mathcal{H}^{\text{pre}} = \mathbb{C}^2$. Put

$$C_{\text{hyp}} z = (z, 0), \quad C_{\text{rep}} z = (z, z), \quad D(a, b) = a.$$

Then

$$DC_{\text{hyp}} z = z = DC_{\text{rep}} z,$$

so the terminal map and its activated range are identical. The second coordinate is repair-only and terminal-null. For

$$Q_\lambda(a, b) := a + \lambda b$$

one nevertheless has

$$Q_\lambda C_{\text{hyp}} z = z, \quad Q_\lambda C_{\text{rep}} z = (1 + \lambda)z.$$

The two pre-terminal output Grams are therefore 1 and $|1 + \lambda|^2$. They can be separated arbitrarily, and the repaired output even vanishes when $\lambda = -1$.

The obstruction in [example 6.7](#) is exactly failure of descent: when $\lambda \neq 0$,

$$\text{Ker } D \not\subseteq \text{Ker } Q_\lambda.$$

Accordingly, repair invariance may be invoked only after proving a common factorization through the literal terminal map,

$$F_X^{\text{hyp}} = A_X L_X^{\text{hyp}}, \quad F_X^{\text{rep}} = A_X L_X^{\text{rep}}, \quad (47)$$

with the same A_X . Equality of terminal norms alone does not imply (47).

Remark 6.8 (L0/L1 claim boundary). The abstract represented-Gram and repair-sandwich consequences are finite-dimensional L0 statements for declared maps and carriers. Identifying the two coordinatewise terminal maps with the literal TPC-64 fixed-shift constructions is an L1 dictionary statement. Any arithmetic use must retain one fixed nonzero shift h_0 , the literal coefficients and terminal multiplier, the actual active support, every required external label until its declared erasure, and the raw counting-measure atomic norm. The exact identities prove neither a favorable activation census nor a polynomial bound for $\mathfrak{J}_X$, $\tau_{X,\nu}$, β_X , or $\Gamma_{X,\nu}$. Finite-scale injectivity can have a superpolynomially small least singular value. Nothing in this section specializes h_0 to 2, controls a parity-sensitive signed kernel, or implies a prime-pair lower bound.

7 Compatible transfer and the one-charge ledger

We now place activation, representation, and coherent reassembly on one carrier. Fix a nonzero parent form $R_{X,\nu}$, the common terminal map L_X with $Y_X \neq \{0\}$, a nonzero represented map $A_X : Y_X \rightarrow H_X$, and a missing/reassembly map $M_X : Y_X \rightarrow H_X$. Set

$$B_X = A_X + M_X.$$

All maps and supports are fixed before z .

Define the exact stage constants

$$\tau_{X,\nu} := \inf_{\mathcal{P}_{X,\nu}(z) > 0} \frac{\|L_X z\|^2}{\mathcal{P}_{X,\nu}(z)}, \quad (48)$$

$$\beta_X := \inf_{0 \neq y \in Y_X} \frac{\|A_X y\|^2}{\|y\|^2}, \quad (49)$$

$$\delta_X := \inf_{A_X y \neq 0} \frac{\|B_X y\|^2}{\|A_X y\|^2}. \quad (50)$$

The direct parent-to-final constant is

$$\Gamma_{X,\nu} := \inf_{\mathcal{P}_{X,\nu}(z) > 0} \frac{\|B_X L_X z\|^2}{\mathcal{P}_{X,\nu}(z)}. \quad (51)$$

The next product is the TPC-62 three-module mechanism specialized to the common activated carrier; its role here is compatibility and loss provenance, not a new abstract multiplication theorem.

Theorem 7.1 (Compatible three-stage certificate). *If $\tau_{X,\nu}, \beta_X, \delta_X$ are defined on the common carrier above, then*

$$\Gamma_{X,\nu} \geq \tau_{X,\nu} \beta_X \delta_X. \quad (52)$$

The product can be sharp, but it need not equal the direct constant and can be arbitrarily smaller.

Proof. If $\mathcal{P}_{X,\nu}(z) > 0$ and $L_X z = 0$, then $\tau_{X,\nu} = 0$ and the claim is trivial. Otherwise apply the three definitions successively to the same vector z , $L_X z$, and $A_X L_X z$. If the latter vanishes, then either $\beta_X = 0$, or the ratio in (50) is outside its nonzero reference set; the displayed bound again reduces to zero. Taking the infimum proves (52). One-dimensional scalar maps give equality. $\square$

The exact reassembly constant δ_X is the TPC-62 shorted generalized eigenvalue relative to $A_X^* A_X$. Its specialization to the current carrier is recorded for compatibility; it is not a new proof of the master short.

Proposition 7.2 (Shorting-collapse dichotomy). *If $\beta_X > 0$, then A_X is injective on Y_X , so the null space in the reassembly short is zero. Consequently*

$$\delta_X = \lambda_{\min} \left[(A_X^* A_X)^{-1/2} B_X^* B_X (A_X^* A_X)^{-1/2} \right]. \quad (53)$$

If the short is genuinely nontrivial, then the uniform staged represented gate has already failed, although the direct final constant may still be positive.

Proof. By [theorem 3.1](#), $\beta_X > 0$ is equivalent to $\text{Ker } A_X = \{0\}$ on Y_X . The nuisance image used in the short is then zero, so the projected Gram reduces to (53). Conversely a nonzero nuisance null space implies $\beta_X = 0$. Direct positivity is logically independent: for example, on $\mathbb{C}^2$, take $A(x_1, x_2) = (x_1, 0)$, $M(x_1, x_2) = (0, x_2)$; then $A + M = I$. $\square$

7.1 Exponent bookkeeping

Suppose, with uniform remainders on the same scope,

$$\tau_{X,\nu} \geq X^{-\lambda_{\text{act}}+o(1)}, \quad \beta_X \geq X^{-\lambda_{\text{rep}}+o(1)}, \quad \delta_X \geq X^{-\lambda_{\text{reasm}}+o(1)}.$$

Then

$$\Gamma_{X,\nu} \geq X^{-(\lambda_{\text{act}}+\lambda_{\text{rep}}+\lambda_{\text{reasm}})+o(1)}. \quad (54)$$

Parent inflation belongs entirely to λ_{act} . The represented Gram is normalized by terminal energy and the reassembly Gram by represented energy, so charging the same repair inflation in either later arrow would be a double count. A downstream arithmetic endpoint still requires the complete loss sum to be *strictly* below its budget; equality at 1/400 is not a certificate.

Nothing in this paper proves any of the three displayed polynomial bounds for the physical unlabelled fixed-shift map. Formula (54) is a compatibility theorem, not a claimed endpoint estimate.

8 Sharp countermodels

The following finite-dimensional models mark the exact limits of the preceding theorems. Some specialize diagnostics already present in TPC-62 to the activated carrier; they are included to close the quantifier audit, not claimed as new counterexamples. None is an arithmetic counterexample.

Example 8.1 (Ambient grouping kernel, positive activated range). *Let $Lz = (z, z) \in \mathbb{C}^2$ and $A(x_1, x_2) = x_1 + x_2$. The ambient map A has kernel $\text{span}(1, -1)$, but this kernel misses $Y = \text{span}(1, 1)$. Since $\|ALz\|^2/\|Lz\|^2 = 2$, the activated constant is 2. An ambient kernel therefore does not decide the range-restricted gate.*

Example 8.2 (Every row visible, global coherent kernel). *Take $T = I_2$ and $F = \begin{bmatrix} 1 & -1 \end{bmatrix}$. Each row column has squared norm 1, but $F(1, 1) = 0$. The normalized Gram*

$$K = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

has eigenvalues 0, 2. Thus even perfect rowwise diagonal data do not give a uniform represented lower bound.

Example 8.3 (Distinguished success, uniform failure). *For $F = \begin{bmatrix} 1 & 1 \end{bmatrix}$, the physical-looking vector $(1, 1)$ has quotient 2, whereas $(1, -1)$ is killed. A favorable distinguished coefficient does not imply a full-space generalized eigenvalue bound.*

Example 8.4 (Injectivity without a polynomial exponent). *Let $T_X = I_2$ and $F_X = \text{diag}(1, e^{-X})$. The map is injective for every finite X , yet $\beta_X = e^{-2X}$. Kernel exclusion alone has no polynomial quantitative content.*

Example 8.5 (Reference route fails while a direct route succeeds). *Let terminal energy be $T = I_2$, choose a reference projection $J(x_1, x_2) = x_1$, and take the final map $A = I_2$. The comparison $\|Jy\|^2/\|y\|^2$ has zero infimum, while the direct A/T constant is 1. Failure of a chosen intermediate reference kills that modular route, not every direct closure.*

Example 8.6 (Stage product can be arbitrarily loose). *For $0 < \varepsilon \leq 1$, on $\mathbb{C}^2$ take parent metric $R = I$, $L = \text{diag}(\sqrt{\varepsilon}, 1)$, and $A = \text{diag}(1, \sqrt{\varepsilon})$, with no reassembly loss. The activation and represented constants are both ε , so their product is ε^2 . But $AL = \sqrt{\varepsilon}I$, and the exact direct constant is ε .*

Example 8.7 (Complementary singular blocks). *Let $T = I_2$, $F_1 = \begin{bmatrix} 1 & 0 \end{bmatrix}$, and $F_2 = \begin{bmatrix} 0 & 1 \end{bmatrix}$. Both block Grams are singular, but the labelled sum is I_2 . This saturates the kernel-fusion criterion. If label erasure instead takes $F_1 = \begin{bmatrix} 1 & 0 \end{bmatrix}$ and $F_2 = \begin{bmatrix} 0 & -1 \end{bmatrix}$ on two independent block inputs, then the coherent map is $(x_1, x_2) \mapsto x_1 - x_2$ and has the diagonal kernel. Labelled fusion and physical fusion are distinct gates.*

Example 8.8 (Terminal-null repair can be read upstream). *Let the repaired parent be $\mathbb{C}^2$, terminalization $L(x, n) = x$, and the original parent be the subspace $n = 0$. Every downstream map AL is repair invariant. The pre-terminal map $F(x, n) = x + Cn$, however, has a Gram whose behavior changes arbitrarily with C . Descent through L is a necessary hypothesis, not notation.*

Example 8.9 (Represented success, reassembly cancellation). *Take $A = I$, a unit vector v , and $M = -vv^*$. The represented constant is 1, but $A + M$ kills v . Activation and represented transfer cannot be used as substitutes for the reassembly short.*

Together these examples show why the actual row Gram, its off-diagonal entries, the label-erasure angle, and the reassembly map must be audited separately.

9 Fixed-shift dictionary, conclusions, and next gate

9.1 What is now exact

On the literal TPC-64 terminal carrier, the following statements are proved without asymptotics:

- (1) the represented reference is the actual activated range Y_X , not a free ambient atom space;
- (2) a genuinely descending represented map is governed by the normalized row Gram (6), with exact kernel and dimension tests;
- (3) every Gram entry is the literal residual-fiber correlation (10);
- (4) terminal-faithful repair leaves downstream Grams unchanged when both descriptions use the same factorization through the common terminal map; it changes direct parent constants only through the parent metric;
- (5) labelled blocks can fuse complementary kernels according to (15), while coherent label erasure requires a separate fusion-angle estimate;
- (6) spectral capacity has a sharp projection law, but row capacity cannot be transported to it without a reducing or basis-transport theorem;

- (7) activation, representation, and reassembly multiply compatibly on the same vector, with each loss charged once.

The prime-resolved construction of Wang [2] gives a conditional benchmark. Under its hypotheses $h_0 \neq 0$, $D = o(J)$, uniqueness of the declared row labels, and sufficiently large X , the refined key $(\gamma, d(m)r, j, p)$ is injective on one declared source system. If this full key and the external block label are retained, the represented map is a coordinate relabelling and hence an isometry on its populated range. In that labelled benchmark $\beta_X^{\text{lab}} = 1$. This does not prove anything about the physical map that forgets p , erases other labels, or coherently reassembles blocks: all unresolved cancellation is localized to those later arrows.

9.2 What remains unproved

This paper does not establish:

- a favorable lower bound for the actual terminal activation ratio $\tau_{X,\nu}$;
- injectivity, let alone a polynomial smallest singular value, for the physical unlabelled represented map;
- a positive physical fusion angle θ_X ;
- a reassembly or downstream lower bound with the literal arithmetic coefficients;
- a statement specialized to $h_0 = 2$, a parity-breaking estimate, or a twin-prime conclusion.

All exact matrix results are L0. The fixed- h_0 identification is L1 only because it preserves the literal carrier, coefficients, and support. An L2 arithmetic advance begins only when one controls the actual matrix $b_{y,m}$ uniformly or on a declared physical cone.

9.3 The next smallest gate

The row formula identifies the next paper’s target. For each active row m , one must determine

- (i) the number and weight of actually exposed output fibers y ;
- (ii) the physical collision degree between the supports of $b_{\cdot,m}$ and $b_{\cdot,n}$;
- (iii) whether the diagonal energy dominates the accumulated off-diagonal correlations on the same fixed-shift carrier.

Support degree alone cannot prove cancellation, but it can turn a subsequent pair-correlation estimate into a Gram bound. This is the route pursued next: actual exposure and collision geometry first, signed cancellation only after the literal neighbor graph is known.

References

- [1] Jr. W. N. Anderson and G. E. Trapp. Shorted operators. II. *SIAM Journal on Applied Mathematics*, 28(1):60–71, 1975.
- [2] Liang Wang. Prime-resolved residual fibers at a fixed shift: Exact grouping SVD, common Möbius gauge, and collision-free transversals, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-55-prime-resolved-residual-fibers. Preprint.
- [3] Liang Wang. Exact closure alternatives for a fixed-shift prime packet: Quantifier-safe gates, shorted-Gram dominance, and localized alias obstructions, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-62-exact-closure-alias-obstructions. Preprint.

- [4] Liang Wang. Canonical cofactor provenance at a fixed shift: Minimal faithful relations, exact exposure factorization, and the single-band completion barrier, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-63-canonical-cofactor-provenance. Preprint.
- [5] Liang Wang. True terminal activation on canonical multiscale incidences: Pair-lift minimality, parent inflation, and sharp capacity laws, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/tpc-64-true-terminal-activation. Preprint.